Math 113 Midterm Practice Problems

Fall 2026 – Worked Solutions

Problem 1

Determine whether each of the following binary algebraic sets is a group, and, if not, which group axiom(s) fail.

(a) $\mathbb{Q}^+$ under division.

Closure	✓	$\frac{a}{b} \in \mathbb{Q}^+$ if $a, b \in \mathbb{Q}^+$
Associativity	✗	$(8/4)/2 = 1 \neq 4 = 8/(4/2)$
Identity	✗	$a/1 = a$ but $1/a \neq a$ for $a \neq 1$
Inverse	✗	no identity

Not a group (associativity, identity and inverses fail).

(b) $\{\sigma \in S_6 \mid \sigma(2) = 4\}$.

Let $H = \{\sigma \in S_6 \mid \sigma(2) = 4\}$.

Closure	✗	for $\sigma = \tau = (2\,4)$: $(\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(4) = 2 \neq 4$
Associativity	✓	composition
Identity	✗	$e(2) = 2 \neq 4$, so $e \notin H$
Inverse	✗	$\sigma = (2\,4\,5) \in H$, $\sigma^{-1}(2) = 5 \neq 4$

Not a group (closure, identity and inverses fail).

(c) $\{\sigma \in S_6 \mid \sigma(2) = 2\}$.

Let $H = \{\sigma \in S_6 \mid \sigma(2) = 2\}$.

Closure	✓	for $\sigma, \tau \in H$: $(\sigma \circ \tau)(2) = \sigma(\tau(2)) = \sigma(2) = 2$
Associativity	✓	composition
Identity	✓	$e(2) = 2$, so $e \in H$
Inverse	✓	$\sigma(2) = 2 \Rightarrow \sigma^{-1}(2) = 2$

A group (it is the stabilizer of 2 in S_6).

(d) $\mathbb{Z}$ under multiplication.

Closure	✓	$a \times b \in \mathbb{Z}$ for $a, b \in \mathbb{Z}$
Associativity	✓	$a \times (b \times c) = (a \times b) \times c$ for $a, b, c \in \mathbb{Z}$
Identity	✓	$a \times 1 = 1 \times a = a$, $1 \in \mathbb{Z}$
Inverse	✗	$a \times b = 1$ forces $b = \frac{1}{a} \notin \mathbb{Z}$ for $a \neq \pm 1$; also if $a = 0$, a^{-1} does not exist

Not a group (inverses fail).

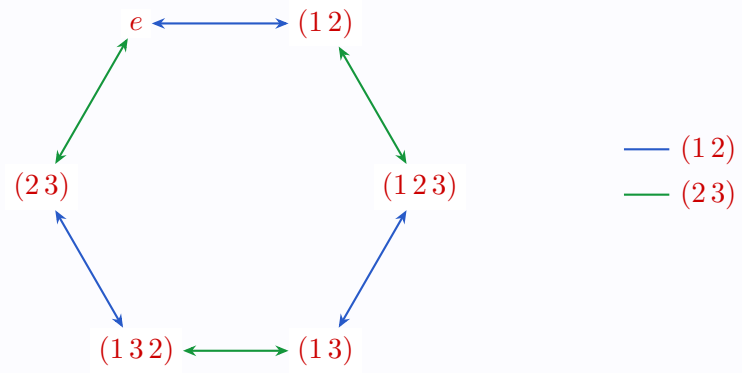
Problem 2

For each of the following generating sets of S_3, draw the corresponding Cayley graph.

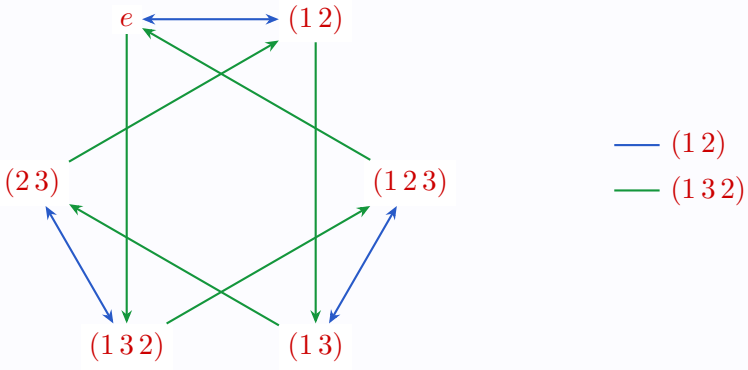
Setup. Permutations compose right-to-left. There is an edge $x \to x \cdot s$ for each generator s; if s has order 2 then $x \cdot s \cdot s = x$, so the edge is drawn with arrows on both ends.

x	$x \cdot (1\,2)$	$x \cdot (1\,3)$	$x \cdot (2\,3)$	$x \cdot (1\,3\,2)$
e	$(1\,2)$	$(1\,3)$	$(2\,3)$	$(1\,3\,2)$
$(1\,2)$	e	$(1\,3\,2)$	$(1\,2\,3)$	$(1\,3)$
$(1\,3)$	$(1\,2\,3)$	e	$(1\,3\,2)$	$(2\,3)$
$(2\,3)$	$(1\,3\,2)$	$(1\,2\,3)$	e	$(1\,2)$
$(1\,2\,3)$	$(1\,3)$	$(2\,3)$	$(1\,2)$	e
$(1\,3\,2)$	$(2\,3)$	$(1\,2)$	$(1\,3)$	$(1\,2\,3)$

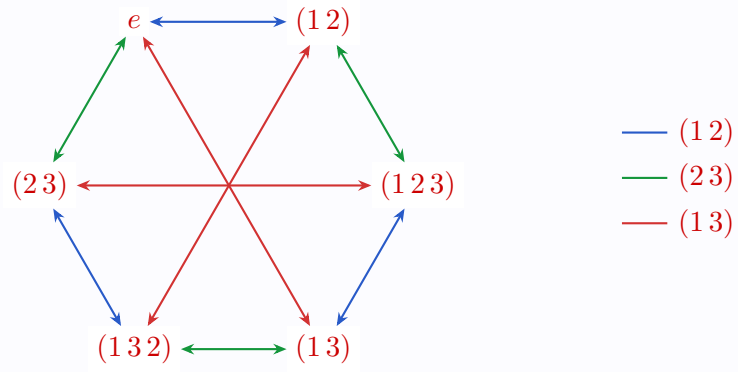
(a) $A = \{(1\,2), (2\,3)\}$



(b) $B = \{(1\,2), (1\,3\,2)\}$



(c) $C = \{(1\,2), (2\,3), (1\,3)\}$



Problem 3

For each of the following groups, decide whether it is abelian, whether it is cyclic, and whether it is simple.

(a) $\mathbb{Z}_{25}$

Abelian	✓	cyclic ⇒ abelian
Cyclic	✓	generator $\langle 1 \rangle$
Simple	✗	25 not prime (an abelian group is simple iff its order is prime; e.g. $\langle 5 \rangle$ is a proper nontrivial normal subgroup)

(b) $\mathbb{Z}_7$

Abelian	✓	cyclic ⇒ abelian
Cyclic	✓	generator $\langle 1 \rangle$
Simple	✓	7 prime

(c) $\mathbb{Z}_2 \times \mathbb{Z}_5$

Abelian	✓	cyclic ⇒ abelian
Cyclic	✓	$\gcd(2,5) = 1$, so $\mathbb{Z}_2 \times \mathbb{Z}_5 \cong \mathbb{Z}_{10}$; generator $\langle (1,1) \rangle$
Simple	✗	10 not prime

(d) S_4

Abelian	✗	$(1\,2)(1\,3) = (1\,3\,2) \neq (1\,2\,3) = (1\,3)(1\,2)$
Cyclic	✗	not abelian
Simple	✗	A_4 is a proper nontrivial subgroup of index 2, hence normal

(e) A_4

Abelian	✗	$(1\,2\,3)(1\,2)(3\,4) = (1\,3\,4) \neq (2\,4\,3) = (1\,2)(3\,4)(1\,2\,3)$
Cyclic	✗	not abelian
Simple	✗	$V_4 = \{e, (1\,2)(3\,4), (1\,3)(2\,4), (1\,4)(2\,3)\}$ is a proper nontrivial normal subgroup of A_4 (conjugation preserves cycle type, so V_4 is closed under conjugation)

(f) A_3

$A_3 = \{e, (1\,2\,3), (1\,3\,2)\} \cong \mathbb{Z}_3$.

Abelian	✓	cyclic $\Rightarrow$ abelian
Cyclic	✓	generator $\langle(1\,2\,3)\rangle$
Simple	✓	3 prime

Problem 4

We say two elements g, h of a group G are *conjugate* (written $g \sim h$) if there exists an element $x \in G$ such that $xgx^{-1} = h$.

(a) Show that $\sim$ is an equivalence relation on G.

Reflexive: $g = ege^{-1}$, so $g \sim g$ (take $x = e$).
Symmetric: Suppose $xgx^{-1} = h$. Then

$$x^{-1}(xgx^{-1})x = x^{-1}(h)(x), \qquad \text{so} \qquad g = x^{-1}hx = x^{-1}h(x^{-1})^{-1}.$$

So $h \sim g$ (using x^{-1}) whenever $g \sim h$.
Transitive: If $h = xgx^{-1}$ and $k = yhy^{-1}$, then

$$k = y(xgx^{-1})y^{-1} = (yx)\,g\,(yx)^{-1}.$$

So $g \sim k$.

(b) Show that if $g \sim h$ and if $\phi : G \to G'$ is a homomorphism, then $\phi(g)$ is conjugate (in G') to $\phi(h)$.

Suppose $xgx^{-1} = h$. Recall $\phi(ab) = \phi(a)\phi(b)$ and $\phi(x^{-1}) = (\phi(x))^{-1}$. Then

$$\phi(h) = \phi(xgx^{-1}) = \phi(x)\,\phi(g)\,\big(\phi(x)\big)^{-1}.$$

Since $\phi(x) = x' \in G'$, we get $\phi(g) \sim \phi(h)$ in G'.

Problem 5

Let G be a group. The normalizer $N_G(S)$ of a subset $S \subseteq G$ is defined to be $\{g \in G \mid gSg^{-1} = S\}$. Show that $N_G(S)$ is a subgroup of G.

Write $N = N_G(S)$.
Identity of G is in N: $eSe^{-1} = S$, so $e \in N$.
Closure: Let $g, h \in N$. Then

$$(gh)S(gh)^{-1} = g(hSh^{-1})g^{-1} = gSg^{-1} = S,$$

so if $g, h \in N$, then $gh \in N$ too.
Inverses: Let $g \in N$, so $gSg^{-1} = S$. Multiplying on the left by g^{-1} and on the right by g:

$$g^{-1}(gSg^{-1})g = g^{-1}Sg \quad \Longrightarrow \quad S = g^{-1}Sg = g^{-1}S(g^{-1})^{-1},$$

so $g^{-1} \in N$ too.
So N is a subgroup of G.

Problem 6

How many abelian groups of order 100 are there, up to isomorphism? List them.

$100 = 2^2 \cdot 5^2$. For each prime there are 2 choices ($\mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$), so there are $2 \cdot 2 = 4$ groups:

- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_5 \times \mathbb{Z}_5$
- $\mathbb{Z}_4 \times \mathbb{Z}_5 \times \mathbb{Z}_5$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_{25}$
- $\mathbb{Z}_4 \times \mathbb{Z}_{25}$

Problem 7

List all elements $a \in \mathbb{Z}_{12}$ such that $\langle a \rangle = \mathbb{Z}_{12}$.

$\langle a \rangle = \mathbb{Z}_{12}$ iff $\gcd(a, 12) = 1$, so $a = 1, 5, 7, 11$.

Problem 8

Suppose $H \leq G$ and H is not normal in G. Explain why we cannot define a group structure on $G/H = \{gH \mid g \in G\}$ as follows: $aH \cdot bH = abH$.

Since H is not normal, there exist $g \in G$ and $h \in H$ with $g^{-1}hg \notin H$. Because $h \in H$, we have $hH = eH$. But

$$(hH)(gH) = hgH \qquad \text{while} \qquad (eH)(gH) = gH,$$

and $hgH = gH \iff g^{-1}hg \in H$, which is false. So the product depends on the choice of coset representatives: the rule is **not well defined**, and $\cdot$ is not a binary operation on G/H.

Problem 9

The action of a group G on a set X is called *free* if for all $x \in X$ and $g \in G$, if $gx = x$ then $g = e$. Faithful and transitive actions are defined in the book. Give examples of group actions with the following properties, or show such an action cannot exist:

(a) free and transitive

✓ Exists. G acting on itself by left multiplication, $g \cdot x = gx$. Free: $gx = x \Rightarrow g = e$. Transitive: for $x, y \in G$, $(yx^{-1}) \cdot x = y$.

(b) faithful and not free

✓ Exists. S_3 acting on $\{1, 2, 3\}$ by $\sigma \cdot i = \sigma(i)$. Faithful: only e fixes $1, 2, 3$. Not free: $(1\,2) \cdot 3 = 3$ but $(1\,2) \neq e$.

(c) free and not faithful

✗ Cannot exist: free $\Rightarrow$ faithful. Suppose $g \cdot x = x$ for all $x \in X$. Pick any $x \in X$ (assuming $X \neq \emptyset$); then $g \cdot x = x$, so freeness gives $g = e$. Hence the only element acting trivially is e, i.e. the action is faithful.

(d) free and not transitive

✓ Exists. $G = \mathbb{Z}_2 = \{e, g\}$ acting on $X = \{1, 2, 3, 4\}$, where g acts as the permutation $(1\,2)(3\,4)$. Free: g moves every point. Not transitive: the orbits are $\{1, 2\}$ and $\{3, 4\}$.

Problem 10

Is there an injective group homomorphism from S_4 to D_{100}?

✗ No. If $\phi : S_4 \to D_{100}$ were injective, then $\phi(S_4) \cong S_4$ would be a subgroup of D_{100} of order 24, so by Lagrange 24 would divide $|D_{100}| = 200$. But $24 \nmid 200$.
Warning: the converse of Lagrange is not true (dividing the order is necessary, not sufficient).

Problem 11

Let $H = \langle (1\,2\,3) \rangle$ denote the cyclic subgroup of A_4 generated by $(1\,2\,3)$. Compute all left cosets of H in A_4 and decide whether H is normal.

$H = \langle (1\,2\,3) \rangle = \{e, (1\,2\,3), (1\,3\,2)\}$, so $|H| = 3$ and $|A_4| = 12$. So there are $\dfrac{|A_4|}{|H|} = \dfrac{12}{3} = 4$ left cosets, each of size $|H| = 3$.

① $eH = \{e, (1\,2\,3), (1\,3\,2)\}$
② $(1\,2)(3\,4)H = \{(1\,2)(3\,4),\ (1\,2)(3\,4)(1\,2\,3),\ (1\,2)(3\,4)(1\,3\,2)\} = \{(1\,2)(3\,4),\ (2\,4\,3),\ (1\,4\,3)\}$
③ $(1\,3)(2\,4)H = \{(1\,3)(2\,4),\ (1\,3)(2\,4)(1\,2\,3),\ (1\,3)(2\,4)(1\,3\,2)\} = \{(1\,3)(2\,4),\ (1\,4\,2),\ (2\,3\,4)\}$
④ $(1\,4)(2\,3)H = \{(1\,4)(2\,3),\ (1\,4)(2\,3)(1\,2\,3),\ (1\,4)(2\,3)(1\,3\,2)\} = \{(1\,4)(2\,3),\ (1\,3\,4),\ (1\,2\,4)\}$

Disjoint ✓ All 12 elements appear once ✓

Not normal ($gH = Hg$ for all $g \in A_4$ fails). Take $g = (1\,4)(2\,3)$:

$$(1\,2\,3)(1\,4)(2\,3) = (1\,4\,2) \in Hg, \qquad (1\,4)(2\,3)(1\,2\,3) = (1\,3\,4) \in gH.$$

But $(1\,4\,2) \notin gH = \{(1\,4)(2\,3), (1\,3\,4), (1\,2\,4)\}$ (coset ④), so $gH \neq Hg$.

Problem 12

The table below defines an action of a group $G = \{g_1, g_2, g_3, g_4, g_5, g_6\}$, with identity element g_1, on the set $X = \{x_1, x_2, x_3, x_4, x_5\}$: the entry in the row labeled x_j and the column labeled g_i is $g_i \cdot x_j$. (You may take for granted that the table does define a valid group action.)

$\cdot$	g_1	g_2	g_3	g_4	g_5	g_6
x_1	x_1	x_2	x_3	x_1	x_2	x_3
x_2	x_2	x_1	x_2	x_3	x_3	x_1
x_3	x_3	x_3	x_1	x_2	x_1	x_2
x_4	x_4	x_5	x_5	x_5	x_4	x_4
x_5	x_5	x_4	x_4	x_4	x_5	x_5

(a) Write down the stabilizer subgroups G_{x_1} and G_{x_4} (Fraleigh calls these the isotropy subgroups).

$G_{x_1} = \{g_1, g_4\}$
$G_{x_4} = \{g_1, g_5, g_6\}$

(b) Find all the orbits of the action.

Orbits: $\{x_1, x_2, x_3\}$ and $\{x_4, x_5\}$.

(c) Verify that $|Gx| \cdot |G_x| = |G|$ both for $x = x_1$ and for $x = x_4$, as predicted by Theorem 16.16 together with Lagrange's theorem. (Here Gx is the orbit of x.)

$|Gx_1| \cdot |G_{x_1}| = 3 \cdot 2 = 6 = |G|$
$|Gx_4| \cdot |G_{x_4}| = 2 \cdot 3 = 6 = |G|$

(d) Determine, with proof, whether the action is: (i) free, meaning that if $g \cdot x = x$ for some $x \in X$ then $g = g_1$; (ii) transitive; (iii) faithful, meaning that the only element $g \in G$ with $g \cdot x = x$ for all $x \in X$ is g_1.

(i) Not free: $g_4 \cdot x_1 = x_1$ but $g_4 \neq g_1$.
(ii) Not transitive: x_1 cannot be sent to x_4 by any g (the orbit of x_1 is $\{x_1, x_2, x_3\}$).
(iii) Faithful: every $g \neq g_1$ moves some point of X:

$$g_2 \cdot x_1 = x_2, \quad g_3 \cdot x_1 = x_3, \quad g_4 \cdot x_2 = x_3, \quad g_5 \cdot x_1 = x_2, \quad g_6 \cdot x_1 = x_3.$$

So the only element fixing every $x \in X$ is g_1 (which acts as the identity).

(e) Using only the table, show that G is not abelian. (Hint: if $g_i g_j = g_j g_i$, then $g_i \cdot (g_j \cdot x) = g_j \cdot (g_i \cdot x)$ for every $x \in X$. Find a specific failure.)

$$g_2 \cdot (g_3 \cdot x_1) = g_2 \cdot x_3 = x_3,$$
$$g_3 \cdot (g_2 \cdot x_1) = g_3 \cdot x_2 = x_2.$$

So $(g_2 g_3) \cdot x_1 = x_3 \neq x_2 = (g_3 g_2) \cdot x_1$, hence $g_2 g_3 \neq g_3 g_2$ and G is not abelian.

(f) Show that G is isomorphic to the symmetric group S_3. (Hint: restrict the action to the orbit of x_1; how many distinct permutations of that orbit appear among the columns of the table?)

Let $O = \{x_1, x_2, x_3\}$ be the orbit of x_1. Since O is an orbit, each $g \in G$ maps O to itself bijectively, giving the permutation $\phi(g) : O \to O$, $x \mapsto g \cdot x$. Reading the columns of the table on O:

g	g_1	g_2	g_3	g_4	g_5	g_6
$\phi(g)$	e	$(x_1\, x_2)$	$(x_1\, x_3)$	$(x_2\, x_3)$	$(x_1\, x_2\, x_3)$	$(x_1\, x_3\, x_2)$

These are 6 distinct permutations of O, i.e. all of $\mathrm{Sym}(O) \cong S_3$.
The map $\phi : G \to \mathrm{Sym}(O)$ is a homomorphism since $(gh) \cdot x = g \cdot (h \cdot x)$, it is injective (the six columns are distinct, so the kernel is trivial), and it is surjective because $|G| = 6 = |\mathrm{Sym}(O)|$. Hence $G \cong S_3$.